



TigerEye

Architectural Spike Report

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1 System Metaphor (Andrea)

System metaphor: “A Bird’s Eye in the Sky”

TigerEye can be described to stakeholders as a “Bird’s Eye in the Sky” — a centralized vantage point that rises above individual systems to provide comprehensive awareness of the threat landscape. From this elevated perspective, operators are no longer confined to fragmented alerts or isolated data sources. The platform will aggregate threat intelligence and network activity into a single, cohesive view which would allow analysts to observe patterns, correlations, and anomalies that would otherwise remain unseen at ground level. Seeing “from the top of the tower”, security teams gain enhanced global visibility across the environment, enabling faster detection, clearer situational understanding, and more confident decision-making. This elevated awareness improves response quality by helping operators prioritize threats, anticipate emerging risks, and coordinate actions with precision — turning reactive security operations into proactive defense.

2 Cycle Intent (Andrea)

Architectural Spike:

The objective of this cycle is to establish a foundational data ingestion and analysis pipeline for the TigerEye platform. This begins with selecting a reputable and reliable cyber-OSINT source and developing a function capable of securely interacting with its API to retrieve relevant threat intelligence data.

In parallel, a baseline storage database will be designed and implemented to support the structured retention of ingested intelligence and network telemetry. This database will serve as the backbone for correlating external threat data with internal observations.

The cycle will focus on aggregating packet capture (PCAP) data generated from the customer’s experimental environment. By centralizing and normalizing this network traffic data, the platform will enable early-stage analysis, enrichment, and correlation with OSINT feeds – laying the groundwork for enhanced visibility and future detection capabilities.

3 Management Plan & Project Management (Aidan & Mihir)

1. Semester Assignments

- a. Architect – Aidan Anderson
 - i. Helping with engineering tasks across different project sectors
 - ii. Designing overall application capabilities based on customer's desired MVP
 - iii. Establishing initial data pipelines from McCrary's testbed environment.
 - iv. Integrating our varied sectors with each other to accomplish various objectives and tasks
- b. Backend Engineer - Amelia Snyder
 - i. Developing the initial parser engine to interpret network traffic
 - ii. Researching and implementing best export format to deliver data to our pattern engine/system
 - iii. Formatting IT/OT network traffic, in a uniform manner.
- c. API integration/Backend Engineer - Mihir Patel
 - i. Researching available sources of OSINT related to both IT and ICS environments
 - ii. Designing a pattern engine to recognize potential threats based on OSINT sources (API integration)
 - iii. Selecting and implementing database solutions for our overall application
- d. Full Stack Engineer - Nick M.
 - i. Working alongside GUI Engineer to select best visualization tools, languages, and products
 - ii. Ensuring accurate data is integrated into threat cases
 - iii. Tagging data with timestamps to help visualize threat timelines for operators
- e. Graphical Interface Engineer - Andrea McCarroll
 - i. Designing the UX/UI for our operators and analyst in mind
 - ii. Researching pre-built infrastructure that could be leveraged to reduce dev time
 - iii. Creating customized dashboards to present our key features such as event timelines, traffic flow diagrams, and incident case queues
- f. Open-Source Engineer - Amy Prieto
 - i. Providing feedback & instruction on architectural decisions to ensure containerization & open-source capability

- ii. Managing the git/code repository
- iii. Collaborating in a full-stack capacity, with back end and front-end engineers on key feature capabilities
- iv. Researching OT cyber domain tools, providing direction on key feature development, and ensuring value to the overall market.

2. Immediate Assignments

a. Aidan:

- i. Research and evaluate Azure and cloud environment options to support TigerEye's architecture.
- ii. Identify required cloud resources, assess feasibility of virtualized servers, and define baseline environment requirements to support data ingestion, OSINT processing, and future scalability.

b. Amelia:

- i. Research the best practices for formatting and exporting security data, including PCAP files and Suricata logs.
- ii. Evaluate structured data formats such as JSON and alternatives and recommend a consistent data formatting approach to support downstream processing, storage, and visualization.

c. Mihir:

- i. Investigate database technologies and architectures suitable for real-time security data ingestion and querying.
- ii. Provide recommendations on database structure (e.g., Elasticsearch) and communicate infrastructure needs to Aidan, including operating system requirements and server specifications.

d. Amy:

- i. Manage GitHub repositories and establish version control best practices for the team.
- ii. Research containerization strategies and identify practices to avoid that could negatively impact performance, maintainability, or collaboration.
- iii. Coordinate with Jason as needed for technical input.

e. Nick / Andrea:

- i. Research front-end technologies and visualization platforms such as React and Grafana.
- ii. Identify effective pre-built visualizations and interaction patterns that support case-based workflows, including alert drilldowns and timeline-based dashboards.

iii. Document UI concepts and recommendations for future integration.

3. Development Schedule

- Parser Engine Beta development: February 6th – February 20th
- OSINT API Beta development: February 6th – February 14th
- Task Check-in: February 11th
- Sponsor Check-in: February 19th or 20th (Based on Sponsor's Availability)
- Stretch goal: Live data pipeline established by February 22nd.
- Retrospective & Schedule assessment – March 4th

4. Meeting Minutes & Cadence

- All meetings begin with a retrospective of last weeks assigned tasks
- Each member reports on task progression, resource needs, and if they're facing any blockers
- At the end of the meeting, each member tells the team their "sweat level" for the upcoming week, so adjustments can be made based on our joint bandwidth.

Every Wednesday – 6-7 PM	Every Thursday – 6-7 PM
Required weekly meeting to discuss progress and go over tasks to be completed before the next meeting.	Optional workshops to follow up technical obstacles and collaborate on solutions.

4 User Stories (Mihir)

- As a security operator, I want the ability to view alerted threats, on a specific device, based on open-source intelligence (OSINT), so that I can better assess defensive countermeasures.
- As a SOC analyst, I want unified IT/OT alerts in a single timeline, so that I can identify cross-domain attack patterns.
- As a cyber security professional and system user, I want to have the ability to set up the system to prioritize alerts and events that my company had determined are our Tier 1 level threats, so that I can better customize the system dashboard to fit my needs.
- As a SOC analyst, I want an interactive, real-time baseline of IT/OT behavior with correlated security events, so that I can identify meaningful deviations, understand incident context, and reduce false positives in industrial environments.

These user stories were collaboratively developed to reflect the core workflow and learning objectives of the TigerEye platform, with each story aligned to support effective threat hunting and situational awareness across IT and OT environments. Together, they emphasize visibility into OSINT-driven device alerts, correlation of IT and OT events within a unified timeline, and the ability to prioritize critical threats based on organizational risk. The final story connects these capabilities through real-time behavioral baselining and interactive visualization, ensuring alerts are contextualized, actionable, and meaningful. Collectively, the user stories define a cohesive workflow that moves from data ingestion and intelligence to correlation, prioritization, and analyst-driven insight.

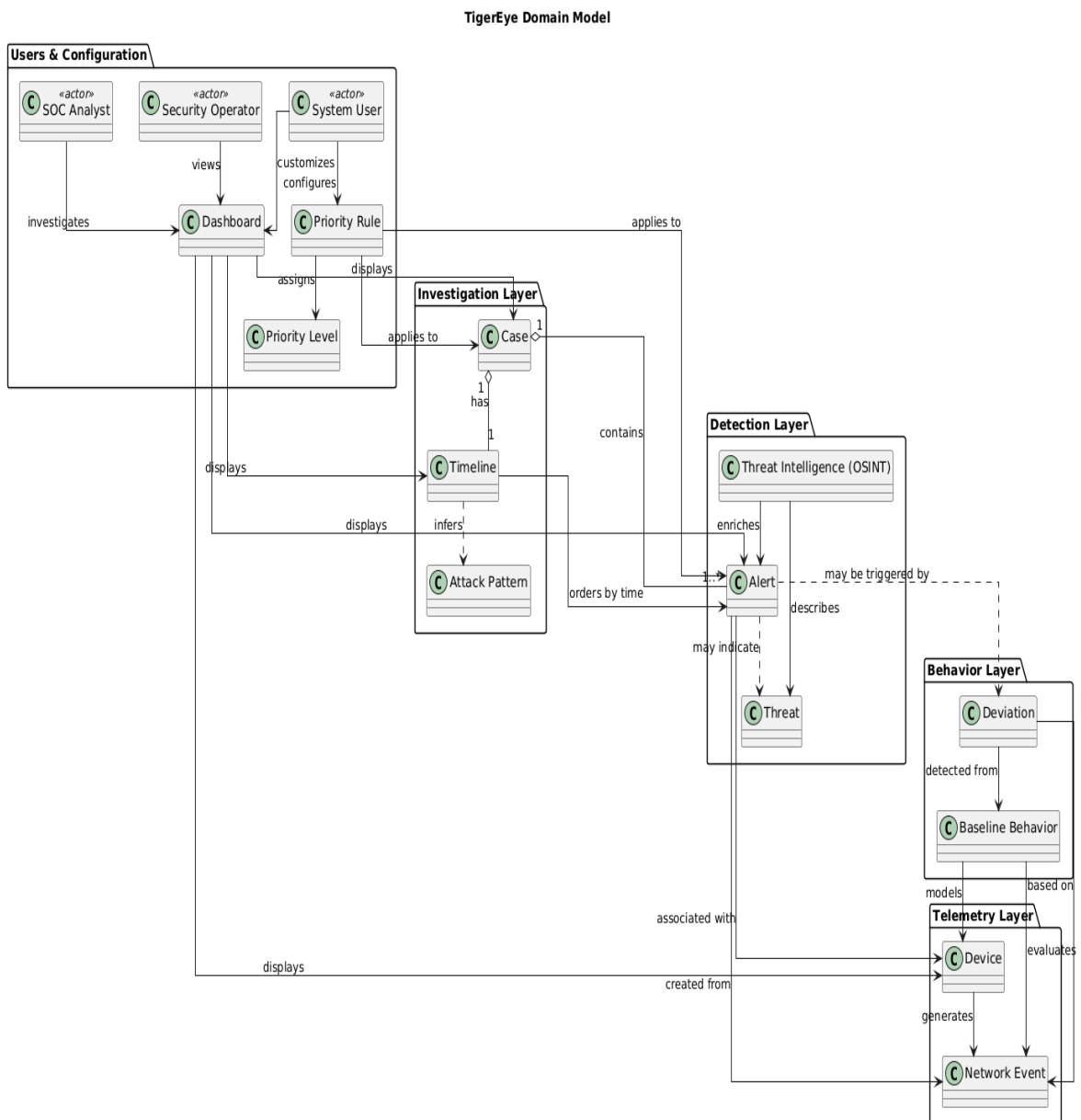
5 Design Documentation (Aidan, Nick & Amelia)

The TigerEye platform will be built on a **Data Pipeline Architecture**, optimized for cross-domain threat hunting by ingesting raw data from IT/OT environments, enriching that information, and presenting it through an interactive interface for rapid analysis. Core architectural components will include:

- **Experimental Environment (Expo):** The pipeline begins by capturing data from endpoints, such as Windows and Linux machines (IT) and SEL Substations (OT), allowing for monitoring of interactions from an Attack Workstation to simulate real-world threats.
- **Backend:**
 - **(Parser Engine):** The initial processing stage that translates raw network traffic from the Expo environment as input, then transforms this data into standardized JSON-formatted logs, ensuring a uniform structure for all subsequent analysis.
 - **(Alert Engine):** The engine correlates the parsed internal logs with that of the external OSINT. External feed utilizes the Open Threat Exchange (OTX) API to ingest global cyber threat data and any specific JSON threat logs. Enriched alert data representing detected threats within the environment will be outputted.
 - **(Database & Storage):** Elasticsearch (under evaluation) will be utilized to store the structured JSON data in a manner suitable for real-time, high-speed, real-time querying of large volumes of network telemetry and security logs.

- **Frontend (TigerEye GUI):** Grafana serves as the interface for the SOC Analyst. The UI will focus on case-based workflows and interactive dashboards. Analysts can perform “drill downs” into specific alerts to view event timelines and technical details.
- **Deployment:** The project will be a Cloud-based solution, utilizing Azure resources provided by the McCrary Institute. The architectural emphasis is on containerization to maintain modularity and support any future transitions to advance the platform.

1. Domain Model



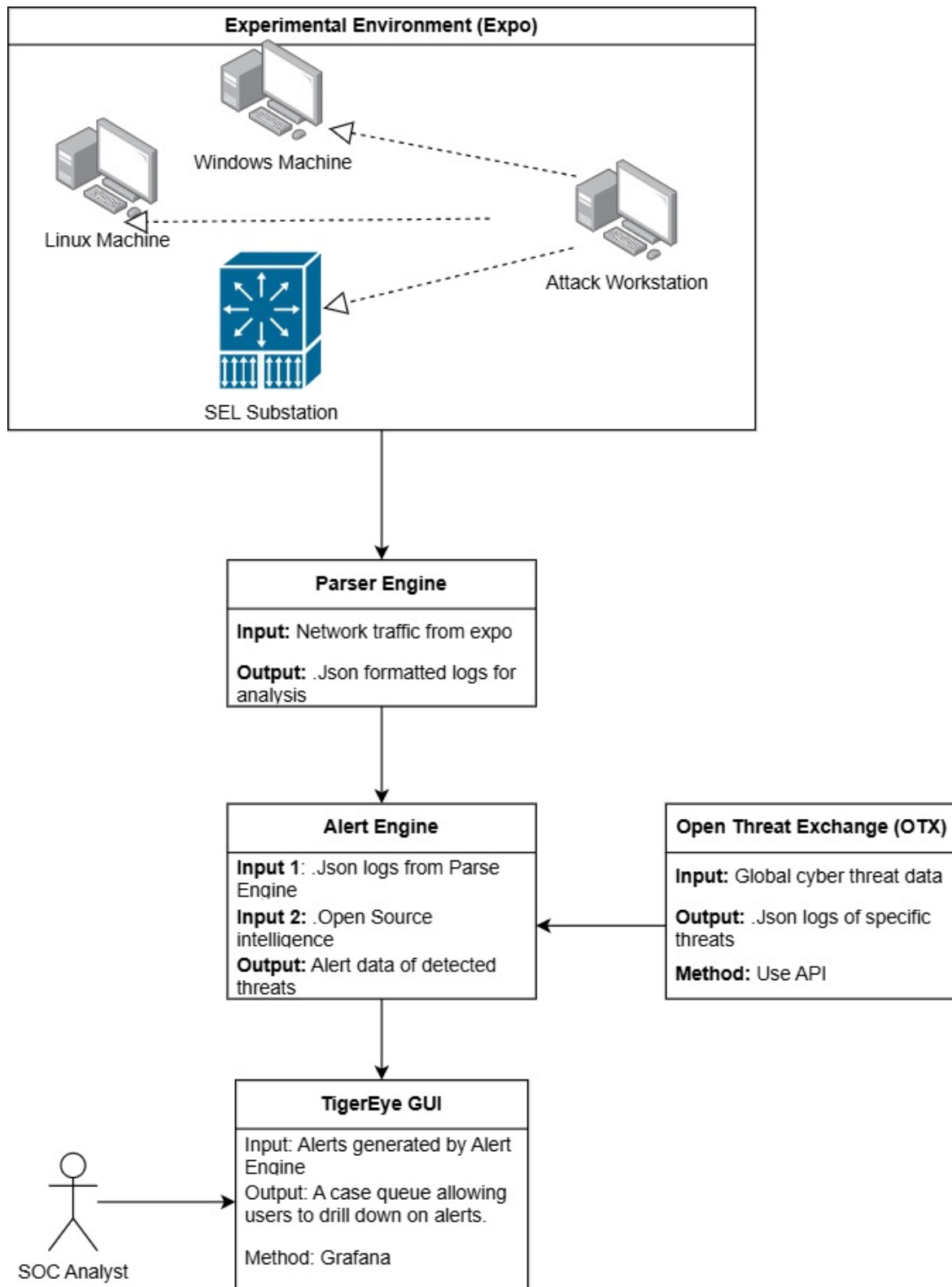
2. Core Entities (Nouns)

- **Device:** An endpoint (IT/OT) within the Expo environment that generates network events.
- **Network Event:** Raw data (Suricata/SEL logs) captured from a Device.
- **Alert:** A security event triggered when a Network Event matches a known threat pattern.
- **Threat Intelligence (OSINT):** External data from OTX used to "enrich" an Alert with context.
- **Case:** A logical grouping of related Alerts assigned for investigation.
- **Timeline:** A chronological sequence of Network Events and Alerts used to infer attack patterns.
- **Priority Rule:** User-defined logic used to assign a **Priority Level** (e.g., Tier 1) to specific threats.

3. Domain Relationships (Logic)

1. **Telemetry Layer:** A **Device** generates multiple **Network Events**.
2. **Detection Layer:** A **Network Event** is evaluated to create an **Alert**, which is then "enriched" by **Threat Intelligence**.
3. **Behavior Layer:** **Network Events** are measured against **Baseline Behavior** to detect **Deviations**.
4. **Investigation Layer:** Multiple **Alerts** are aggregated into a **Case**, which has a corresponding **Timeline** to help analysts infer an **Attack Pattern**.
5. **Configuration Layer:** A **System User** defines **Priority Rules** that assign **Priority Levels** to specific **Dashboards** and **Cases**.

Conceptual Diagram



6 Risk Mitigation (Aidan)

Risk	Description	Mitigation
Limited tools for interpreting ICS network protocols	Operational technologies tend to have vendor specific net protocols that have limited open-source tools. This could increase our development time overall, due to us having to create specialized classes/functions to parse OT data from scratch.	Our primary mitigation is to use .json as our overall data format. This way we can build a parse engine to take in raw packet capture data, tag interactions with our metadata, and funnel it into our pipeline. Versus, relying on other libraries that might not support industrial protocols.
Lack of reliable OSINT resources with functional APIs	Due to the confidentiality of threats against critical infrastructure, most data sources for traditional artifacts of cyber intelligence (CVEs, IoCs, TTPs) are limited to those with federal connections to CISA. Which creates an intelligence bottle neck for open-source options. Additionally, majority of these threat exchanges do not have federal funding or high commercial investment, so their APIs might not be as refined as mainstream providers such as Mandiant Advantage or Microsoft threat intelligence.	The primary mitigation for under sourced intelligence, is to begin developing your own. From a design perspective, we would shift our focus to verifying data to provide from these sources, instead of using them to inform our platform.

Proof of User Stories approval:

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TigerEye Sponsor Check-in

Summarize

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Best,
Aidan



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Craig Whittinghill

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To: ✓ Aidan Anderson

Cc: 🕒 Nicholas Mcgallagher; ✓ Mihir Patel; 🕒 Amy Prieto; ✓ Andrea McCarroll; ✓ Amelia Snyder

Wed 1/28/2026 4:14 PM

Aidan, thanks for sharing your user stories!

Those look like really good ones. I am excited to see more as you progress. Think you're on a good track!

Craig

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